

APPENDIX

Part II: A simple yet effective disequilibrium model of the Beveridge curve

D.

D.1. Data analysis

We compare three search-and-matching labor market models, all written in terms of labour-force *fractions* ($0 < u, v < 1$) but displayed below in percentage points for readability. Throughout we work with monthly U.S. data¹ spanning January 1951 – December 2023.

- Vacancies are floored at 10^{-4} to avoid division by zero.
- GDP is filtered using a HP filter allowing to extract the cycle component from the trend and then *standardized with a 120-month rolling standard deviation*, then lagged one months (g_{t-1} drives $\dot{u}_t$).²
- We assume that, for Model 1, the spontaneous separation probability s is equal to a fraction of the empirical average separation probability measured by Hobijn and Sahin ($s_0 = 0.0106$), found through the following process: we compute

$$s' = \frac{1}{N} \sum_{t=1}^N \left(s_0 + C \left[\frac{dg}{dt} \right]_- \right),$$

where N is the number of observations and $\left[\frac{dg}{dt} \right]_-$ denotes the negative part of the derivative of GDP. We then set $s \leftarrow s'$ and recompute the average to obtain

$$s'' = \frac{1}{N} \sum_{t=1}^N \left(s' + C \left[\frac{dg}{dt} \right]_- \right).$$

Finally, we update the target value with

$$s \leftarrow s'' - s'.$$

This process converges to $s = 0.00265$. Models 2–3 treat s as free.

D.2. Fitting procedure and error evaluation for unemployment

We estimate each model by non linear least squares (NLLS), holding the separation rate fixed at $s = 0.00265$. For Models 1 and 3, which feature state-dependent dynamics, we define a residual vector

$$r_i(\theta) = u_{\text{sim},i}(\theta) - u_i, \quad i = 1, \dots, N,$$

where $u_{\text{sim},i}(\theta)$ is obtained by iterating $\Delta u_t = f(u_{t-1}, v_{t-1}, \Delta g_{t-1}; \theta)$. For Model 2, the static root form $u_t = s/(s + K v_t)$ is likewise recast as $r_i(K) = s/(s + K v_i) - u_i$. We then call `scipy.optimize.least_squares` with a finite-difference Jacobian. At the optimum $\hat{\theta}$, the estimated co-

1: JOLTS vacancy rate from 2001 aggregated with composite Help Wanted Index data spanning from 1951, BLS unemployment rate, and FRED real GDP converted to quarterly cyclical component by one-sided HP filter, then linearly interpolated to months.

2: We empirically find that a lag of one months gives the best evaluation metric when fitting the data.

variance matrix is

$$\widehat{\text{Var}}(\hat{\theta}) = \hat{\sigma}^2 [J(\hat{\theta})^\top J(\hat{\theta})]^{-1}, \quad \hat{\sigma}^2 = \frac{\text{SSE}(\hat{\theta})}{N - p},$$

where J is the $N \times p$ Jacobian of residuals, p is the number of parameters and SSE denotes the sum of squared errors. The diagonal entries of $\widehat{\text{Var}}(\hat{\theta})$ yield standard errors, and we form two-sided 95 % confidence intervals via $\hat{\theta}_j \pm t_{0.975, N-p} \text{se}(\hat{\theta}_j)$.

To test whether Model 1's "shock" parameter C is significantly different from zero, we note that Model 3 is nested within Model 1 via $C = 0$. Denoting SSE_{full} and SSE_{rest} as the residual sums of squares of Models 1 and 3 respectively, the classical F -statistic

$$F = \frac{(\text{SSE}_{\text{rest}} - \text{SSE}_{\text{full}})/1}{\text{SSE}_{\text{full}}/(N - 2)} \sim F(1, N - 2)$$

evaluates the null $C = 0$. Finally, because Model 2 and Model 1 are not nested, we carry out a residual bootstrap with $B = 500$ replications. For each draw, we resample the original model's residuals with replacement, re-fit both Model 1 and Model 2 to the pseudo-data, and record the bootstrap distributions of R^2 . We then compute 95 % bootstrap confidence intervals for $R^2_{(1)}$ and $R^2_{(2)}$, as well as for $\Delta R^2 = R^2_{(1)} - R^2_{(2)}$. Because the 95 % CI for ΔR^2 does not cross zero, we conclude that Model 1's in-sample fit is significantly better than Model 2's at the 5 % level.

Model	Param	Estimate	Std. Err.	95% CI Lower	95% CI Upper
1	C	0.011998	0.000631	0.010760	0.013237
	K	1.730775	0.025917	1.679908	1.781643
2	K	1.397351	0.014049	1.369778	1.424924
3	K	1.343137	0.014484	1.314710	1.371563

Table D.1.: Estimated Parameters (s fixed at 0.00265)

Model	R^2	MAE	RMS	Corr.
1	0.3733	0.01113	0.01362	0.6537
2	0.1265	0.01245	0.01608	0.5282
3	0.0600	0.01343	0.01668	0.3895

Table D.2.: Fit-Quality Metrics

Quantity	Value
SSE(Model 3, restricted)	0.243587
SSE(Model 1, full)	0.162389
$F(1, 874)$	437.017
p -value	0.0000
Decision	Reject $H_0: C = 0$. Model 1 fits significantly better than Model 3.

Table D.3.: F-Test for Model 1 vs Model 3 ($C = 0$ Restriction)

Description	Value
Bootstrap replicates (B)	500
Elapsed time (sec)	7.6
Model 1: R^2	0.3733 (95% CI: [0.3253, 0.4280])
Model 2: R^2	0.1265 (95% CI: [0.0326, 0.2371])
$\Delta R^2 = R^2_{(1)} - R^2_{(2)}$	0.2469 (95% CI: [0.1517, 0.3390])
Fraction of $\Delta R^2_{\text{boot}} > 0$	1.0000
Inference	Model 1's R^2 is significantly greater than Model 2's at the 5% level (no overlap).

Table D.4.: Bootstrap Inference for R^2 (Model 1 vs Model 2)

Results for vacancies

The results for fitting vacancies, when letting all parameters be free, can be summarized in the following tables:

Parameter	Estimate
v_0	0.005
C_v	0.008
K	2.65

Table D.5.: Estimated parameters for the vacancy equation

Metric	Value
R^2	0.09
MAE	0.007
RMS	0.01
Pearson correlation	0.52

Table D.6.: Goodness-of-fit metrics

D.3. Additional plots

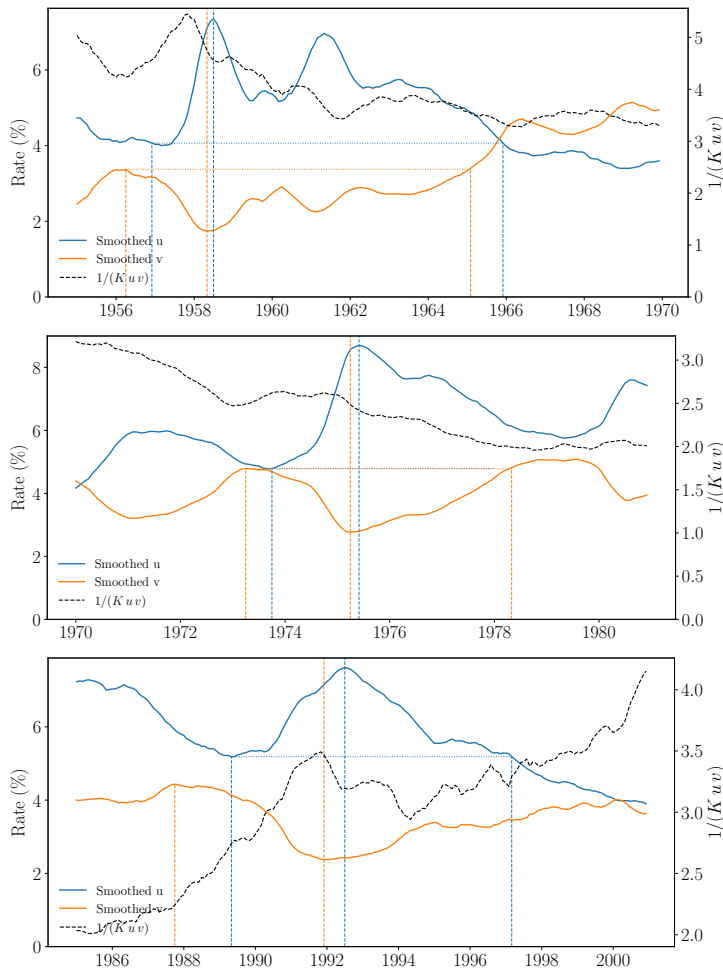


Figure D.1: *Asymmetric lags for several periods recession.* Smoothed monthly unemployment (blue) and vacancies (orange) for the United States for several time periods, with the characteristic matching time $1/(Kuv)$ (black, right-hand scale) plotted in units of months required to achieve a one percent change based on matching alone. Vertical dashed lines represent key events. We overlay dashed colored lines to distinguish the mean slopes of unemployment and vacancy rates during recessionary and recovery phases. Our analysis shows that, for both series and all periods, the average slope during recessions is steeper than the subsequent recovery slopes.

Titre : Modèles minimaux hors-équilibre de systèmes socio-économiques : du marché immobilier français au marché du travail américain.

Mots clés : Physique statistique, hors équilibre, systèmes socio-économiques, modèles d'agents.

Cette thèse étudie la dynamique complexe et souvent contre-intuitive des systèmes socio-économiques en mariant la rigueur mathématique de la physique statistique à la flexibilité adaptative des simulations de modèles d'agents. À travers deux volets complémentaires, nous montrons que des interactions purement locales — qu'il s'agisse de champs continus comme les prix de l'immobilier ou d'agents individuels suivant des règles de décision simples et limitées — donnent naissance à des phénomènes agrégés riches que les théories économiques classiques ne peuvent pas expliquer. Dans la première partie, nous revisitons le modèle de ségrégation de Sakoda-Schelling sur un réseau. À partir d'un modèle d'agents qui évaluent localement la densité perçue, nous réalisons une analyse de stabilité linéaire et des simulations à grande échelle pour en déduire des équations hydrodynamiques stochastiques, nous permettant de lier le comportement hors d'équilibre du modèle à la théorie de la matière active. Nous démontrons que les fonctions d'utilité non linéaires suffisent à engendrer une transition de phase, pour une large gamme de paramètres, appartenant à la classe d'universalité du modèle d'Ising en 2D. Nous étudions ensuite le marché immobilier français via une équation de diffusion stochastique micro-fondée heuristiquement à travers un ajustement de prix local et du bruit microscopique. Calibrée sur un demi-siècle de données régionales, cette équation reproduit les régularités spatiales et temporelles observées — variogrammes spatiaux log-

croissants, autocorrélations longue portée, persistance des chocs sur plusieurs années —, et une analyse par réponse impulsionnelle fournit une constante de diffusion vraisemblable, confirmant des conjectures de longue date et soulignant le caractère non efficient de ce marché.

La seconde partie étudie un modèle d'agents zéro-intelligence dans lequel des demandeurs d'emploi postulent aléatoirement à des offres compatibles avec leurs compétences, tandis que les entreprises ajustent leurs annonces selon des règles heuristiques simples. De ces simulations discrètes, nous établissons rigoureusement des équations différentielles qui modélisent les délais de coordination, l'asymétrie entre embauches et licenciements, et le processus de mise en relation non instantané. Ces lois continues comblent le fossé micro-macro et montrent comment des retards purement mécaniques génèrent l'hystérésis observée sur la courbe de Beveridge, sans faire appel aux hypothèses classiques d'agents rationnels. Nous en extrayons enfin un modèle dynamique minimal de déséquilibre, calibré sur les données américaines de 1951 à 2023. À partir d'une trajectoire de taux de place vacantes donnée, ce modèle parcimonieux reproduit fidèlement le chômage observé. Son caractère analytique le rend particulièrement utile pour l'analyse de politiques publiques, en capturant la mécanique hors-équilibre essentielle des fluctuations du marché du travail.

Title : Minimal Non-Equilibrium Models of Socio-Economic Systems: From the French Housing Market to the US Labor Market.

Keywords : Statistical Physics, Out-of-Equilibrium, Socio-Economic Systems, Agent-Based Models.

This thesis investigates the complex, often counter-intuitive dynamics of socio-economic systems by fusing the mathematical rigor of statistical physics with the adaptive flexibility of agent-based simulations. Across two complementary strands, we show that purely local interactions—whether governing continuous fields such as housing prices or guiding individual agents through simple, boundedly rational decision rules—generate rich aggregate phenomena that classical equilibrium paradigms cannot explain. In the first part of this thesis, we revisit the Sakoda-Schelling segregation model on a “neighborhood-less” lattice. From an agent-based model where each agent evaluates only the density in the vicinity of its own site, we perform linear-stability analysis and large-scale simulations to derive stochastic hydrodynamic equations mapping its out-of-equilibrium dynamics onto active-matter form. We demonstrate that nonlinear utilities alone drive phase-separation—segregation patterns—over a broad parameter range, and we identify the transition as belonging to the 2D Ising universality class. We then turn to the French housing market. We propose a stochastic diffusion equation micro-founded on simple, locally applied price-adjustment heuristics and microscopic noise. Calibrated on fifty years of regional price data, this equation reproduces key spatial and tempo-

ral regularities—variograms growing logarithmically with distance, long-range autocorrelations and multi-year shock persistence. An impulse-response analysis yields a diffusion constant of plausible magnitude, confirming longstanding conjectures about price diffusion and highlighting deviations from efficient-market behavior.

The second part of this thesis studies a zero-intelligence agent-based model in which unemployed workers apply randomly to skill-compatible vacancies and firms adjust postings via simple heuristics. From these discrete simulations we rigorously derive continuous disequilibrium equations isolating coordination lags, asymmetric hiring versus firing speeds and non-instantaneous matching. These laws bridge the micro-macro gap and show how purely mechanical delays generate the observed Beveridge-curve hysteresis, without rational-expectations assumptions. We finally distill these insights into a minimal dynamical disequilibrium model: a small system of coupled differential equations calibrated on U.S. data (1951–2023). Given an observed vacancy trajectory, this parsimonious model reproduces the unemployment path with high fidelity. Its analytical tractability makes it a powerful tool for policy analysis, capturing the essential out-of-steady-state mechanics of labor-market fluctuations.